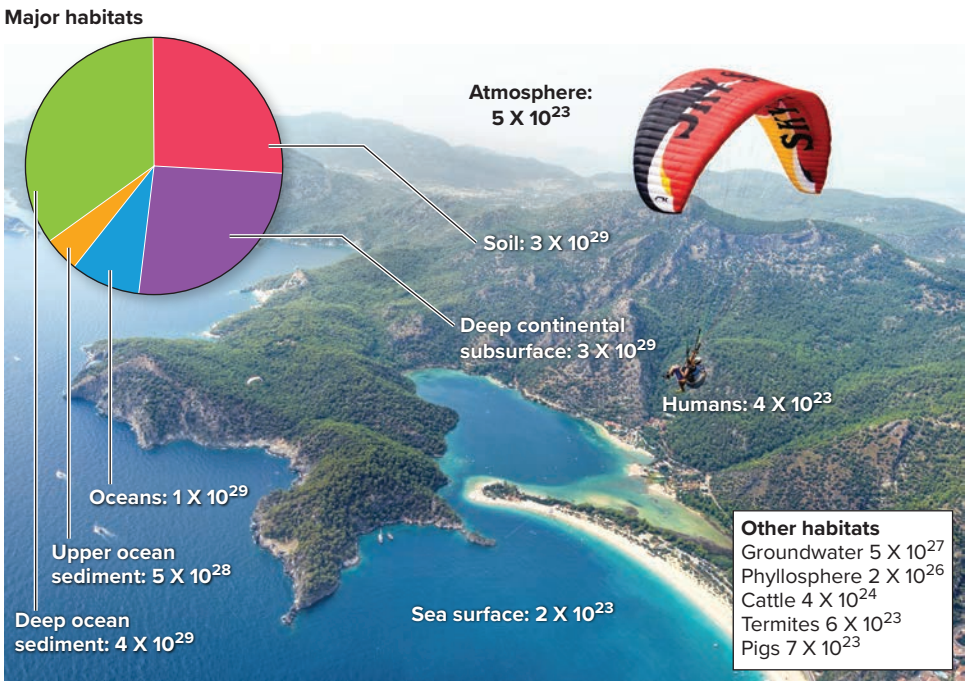


Figure 1.2 Bacterial and Archaeal Habitats and Abundance. Numbers indicate the number of microbial cells in each habitat. The majority of bacteria and archaea live in oceans and sediments, either within the Earth's crust or deep below the crust (subsurface). The discovery of viable microbes so deep within our planet is a recent and exciting development. Other habitats include the phyllopshere (above ground portions of plants), livestock, and humans.

Fotout/Shutterstock

many similarities and differences. Third, the genomic revolution enabled the analysis of nucleic acid and protein sequences from a wide variety of organisms. The comparison of ribosomal RNA (rRNA) nucleic acid sequences, begun by Carl Woese (1928–2012) in the 1970s, transformed our understanding of the term *prokaryote*. It was discovered that there are two very different groups of organisms with prokaryotic cell morphology: Bacteria and Archaea. Among eukaryotic microbes, later studies showed that Protista is not a cohesive taxonomic unit (i.e., taxon) and that it should be divided into three or more kingdoms. These studies and others led many taxonomists to reject the five-kingdom system in favor of one that divides cellular organisms into three domains: Bacteria, Archaea, and Eukarya (all eukaryotic organisms) (**figure 1.3**). ▶| *Nucleic acids (appendix I); Proteins (appendix I)*

Members of domain **Bacteria** are usually single-celled organisms.[1] Most have cell walls that contain the structural molecule peptidoglycan. Despite popular belief, most bacteria do not cause disease. In fact, bacteria are major inhabitants of our bodies, forming the human **microbiome.** Indeed, at least as many microbial cells are found in and on the human body as there are human cells. These microbes contribute to the development of the body's immune system. Microbes that inhabit the large intestine help the body digest food and produce vitamins. In these and many other ways, the human microbiome helps maintain our health and well-being. ▶| *Overview of bacterial cell wall structure (section 3.4); Human microbiome and host interactions (chapter 33)*

Unfortunately some bacteria do cause disease, and some of these diseases can have a huge impact. In 1347 the plague (Black Death), a disease caused by bacteria living in fleas, struck Europe with brutal force, killing one-third of the population within 4 years. Over the next 80 years, the disease struck repeatedly, eventually wiping out roughly half of the European population. The resulting labor shortage gave workers more power, ultimately eliminating serfdom, and preparing the way for the Renaissance.

Members of domain **Archaea** are distinguished from bacteria by many features, most notably their distinctive rRNA sequences, cell walls, and membrane lipids. Some have unique metabolic characteristics, such as the ability to generate methane (natural) gas. Some archaea are found in extreme environments, including those with high temperatures (thermophiles) and high concentrations of salt (extreme halophiles). Archaea do not appear to directly cause disease in humans.

Domain **Eukarya** includes plants, animals, and microorganisms classified as protists or fungi. **Protists** are generally unicellular but larger than most bacteria and archaea. They have traditionally been divided into **protozoa,** which have an animal-like metabolism, and **algae,** which are photosynthetic. However, these terms lack taxonomic value because protists, algae, and protozoa do not form three groups, each with a single evolutionary history. Nonetheless, for convenience, we use these terms here. ▶| *Protist diversity reflects broad phylogeny (section 23.1)*

Fungi are a diverse group of microorganisms that range from unicellular forms (yeasts) to multicellular molds and mushrooms. Because of their metabolic capabilities, many fungi play beneficial roles, including making bread dough rise, producing antibiotics, and decomposing dead organisms. Some fungi associate with plant roots to form mycorrhizae. Mycorrhizal fungi transfer nutrients to the roots, improving growth of the plants, especially in poor soils. Other fungi cause plant diseases (e.g., rusts, powdery mildews, and smuts) and diseases in humans and other animals. ▶| *Fungal biology reflects vast diversity (section 24.1)*

1 In this text, the term *bacteria* (s., *bacterium*) is used to refer to those microbes belonging to domain Bacteria, and the term *archaea* (s., *archaeon*) is used to refer to those that belong to domain Archaea. In some publications, the term *bacteria* is used to refer to all cells having prokaryotic cell structure. That is not the case in this text.

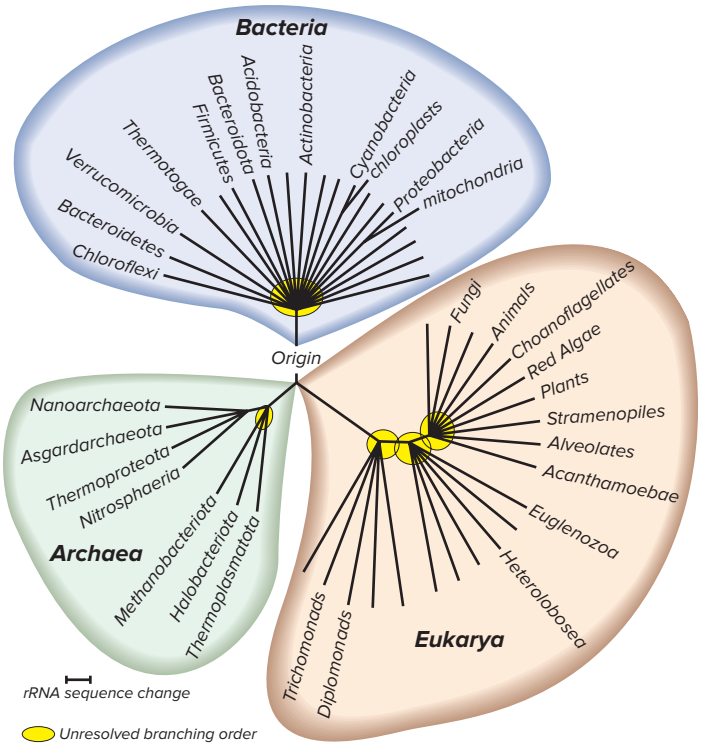


Figure 1.3 Universal Phylogenetic Tree. Only representative lineages have been identified.

MICRO INQUIRY *How many of the taxa listed in the figure include microbes?*

The microbial world also includes numerous acellular infectious agents. **Viruses** are acellular entities that must invade a host cell to multiply. The simplest virus particles (also called virions) are composed only of proteins and a nucleic acid, and can be extremely small (the smallest is 10,000 times smaller than a typical bacterium). However, their small size belies their power. They cause many animal and plant diseases and, as we saw most recently with COVID-19, can trigger epidemics and pandemics that shape human history. In addition to COVID-19, viral diseases include rabies, influenza, AIDS, the common cold, and some cancers. Viruses are also important in aquatic environments, where they play a critical role in shaping microbial communities. **Viroids** are infectious agents composed only of ribonucleic acid (RNA). They cause numerous plant diseases. **Satellites** are composed of a nucleic acid enclosed in a protein shell. They must coinfect a host cell with a virus, called a helper virus, to complete their life cycle. Satellites and their helper viruses cause both plant and animal diseases. Finally, **prions,** infectious agents composed only of protein, are responsible for causing neurological diseases such as scrapie and "mad cow disease." ▶| *Viruses and other acellular infectious agents (chapter 25)*

Comprehension Check

1. How did the methods used to classify microbes change, particularly in the last half of the twentieth century? What was the result of these technological advances?
2. Identify one characteristic for each of these types of microbes that distinguishes it from the other types: bacteria, archaea, protists, fungi, viruses, viroids, satellites, and prions.
3. Describe one interaction with microbes you had yesterday.

1.2 Microbes Have Evolved and Diversified for Billions of Years

After reading this section, you should be able to:

- **a.** Explain the RNA world hypothesis and the evidence that supports it
- **b.** Design a set of experiments that could be used to place a newly discovered cellular microbe on a phylogenetic tree based on small subunit (SSU) rRNA sequences
- **c.** Compare and contrast the evolution of mitochondria and chloroplasts

A review of figure 1.3 reminds us that microbes are the dominant organisms on Earth. How has microbial life been able to radiate to such an astonishing level of diversity? To answer this question, we must consider microbial evolution. The field of microbial evolution, like any other scientific endeavor, is based on the formulation of hypotheses, the gathering and analysis of data, and the reformation of hypotheses based on newly acquired evidence. That is to say, the study of microbial evolution is based on the scientific method. To be sure, it is difficult to amass evidence when considering events that occurred millions, and often billions, of years ago, but the application of molecular methods has revealed a living record of life's ancient history. This section describes the outcome of this scientific research.

Theories of the Origin of Life Depend Primarily on Indirect Evidence

Dating meteorites through the use of radioisotopes places our planet at an estimated 4.5 to 4.6 billion years old. However, conditions on Earth for the first 100 million years or so were far too harsh to sustain any type of life. Eventually bombardment by meteorites decreased, water appeared on the planet in liquid form, and gases were released by geological activity to form Earth's atmosphere. These conditions were amenable to the origin of the first life forms. But how did this occur, and what did these life forms look like?

To find evidence of life and develop hypotheses about its origin and subsequent evolution, scientists must be able to define life. Although even very young children can examine an object

and correctly determine whether it is living or not, defining life succinctly is actually not that easy. Most definitions of life consist of a set of attributes. The attributes of particular importance to paleobiologists are an orderly structure, the ability to obtain and use energy (i.e., metabolism), and the ability to reproduce. Just as NASA scientists are using the characteristics of microbes on Earth today to search for life elsewhere, so too are scientists examining **extant organisms,** those organisms present today, to explore the origin of life. Some extant organisms have structures and molecules that represent relics of ancient life forms. These can provide scientists with ideas about the type of evidence to seek when testing hypotheses.

The best direct evidence for the nature of primitive life would be a fossil record. There have been reports of microbial fossil discoveries since 1977. These have always met with skepticism because some objects that look like cells can be formed by geological forces that occurred as the rock was formed. The result is that the fossil record for microbes is sparse and always open to reinterpretation.

Despite these problems, most scientists agree that life was present on Earth about 3.5 to 3.8 billion years ago (**figure 1.4**). To reach this conclusion, biologists rely on indirect evidence. Among the indirect evidence used are molecular fossils. These are chemicals found in rock or sediment that are chemically related to biological molecules. For instance, the presence of molecules called hopanes in a rock indicates that bacteria were present when the rock was formed. This conclusion is reached because hopanes are formed from hopanoids, which are found in the plasma membranes of extant bacteria. As you can see, no single piece of evidence can stand alone. Instead many pieces of evidence are put together in an attempt to get a coherent picture to emerge, as with a jigsaw puzzle.

Early Life Was Probably RNA-based

Before there was life, most evidence suggests that Earth was a very different place: hot and anoxic, with an atmosphere rich in water vapor, carbon dioxide, and nitrogen. In the oceans, hydrogen, methane, and carboxylic acids were formed by geological and chemical processes. Areas near hydrothermal vents may have provided the conditions that allowed chemicals to react with one another, randomly testing the usefulness of the reaction and the stability of its products. Some reactions generated molecules that functioned as catalysts, some aggregated with other molecules to form the predecessors of modern cell structures, and others were able to replicate and act as units of hereditary information (**Microbial Diversity & Ecology 1.1**).

How did early cells, sometimes called *probionts*, arise? In modern cells, three different molecules fulfill the roles of catalysts, structural molecules, and hereditary molecules. Proteins have two major roles in modern cells: catalytic and structural. Catalytic proteins are **enzymes** and structural proteins serve myriad functions, such as transport, attachment, and motility. DNA stores hereditary information that is replicated and passed on to the next generation. RNA converts the information stored in DNA into protein. Any hypothesis about the origin of life must account for the evolution of these molecules, but their relationships to each other in modern cells complicates attempts to imagine how they evolved. Proteins can do cellular work, but their synthesis involves other proteins and RNA, and uses information stored in DNA. DNA cannot do cellular work, and proteins are needed for its replication. RNA synthesis requires both DNA as the template and proteins as catalysts.

Based on these considerations, it is hypothesized that at some time in the evolution of probionts, there must have been a single molecule that could do both cellular work and replicate. This idea was supported in 1981 when Thomas Cech discovered an RNA molecule in a protist (*Tetrahymena* sp.) that also had catalytic activity. Since then, other catalytic RNA molecules have been discovered, including an RNA found in ribosomes that is responsible for forming peptide bonds—the bonds that hold together amino acids, the building blocks of proteins. Catalytic RNA molecules are now called **ribozymes.**

The discovery of ribozymes suggested that RNA at some time was capable of storing, copying, and expressing genetic information, as well as catalyzing other chemical reactions. In 1986 Nobel laureate Walter Gilbert coined the term **RNA world** to describe this precellular stage in the evolution of life. However, for this precellular RNA-based stage to proceed to the evolution of cellular life forms, a lipid membrane must have formed around the RNA (**figure 1.5**). This important evolutionary step is easier to imagine than other events in the origin of cellular life forms because lipids, major structural components of the membranes of modern organisms, spontaneously form liposomes—vesicles bounded by a lipid bilayer. The notion of an RNA world has caused some scientists to look for evidence on Mars, where conditions are thought to have been frozen in the prebiotic era.

▶| *Lipids (appendix I)*

Back here on Earth, Jack Szostak, also a Noble laureate, is a leader in experimentally simulating how protobionts containing only RNA may have formed. When his group created liposomes using simpler fatty acids than those found in membranes today, the liposomes were leaky. These leaky liposomes allowed single RNA nucleotides to move into the liposome, but prevented large RNA chains from moving out. Furthermore, researchers could prod the liposomes into growing and dividing. Dr. Szostak's group has also been able to create conditions in which an RNA molecule could serve as a template for synthesis of a complementary RNA strand. These experiments may have recapitulated early steps in the evolution of cells. As seen in figure 1.5, several other processes need to occur to reach the level of complexity found in extant cells.

Apart from its ability to perform catalytic activities, the function of RNA suggests its ancient origin. Consider that much of the cellular pool of RNA in modern cells exists in the ribosome, a structure that consists largely of ribosomal RNA (rRNA) and uses messenger RNA (mRNA) and transfer RNA

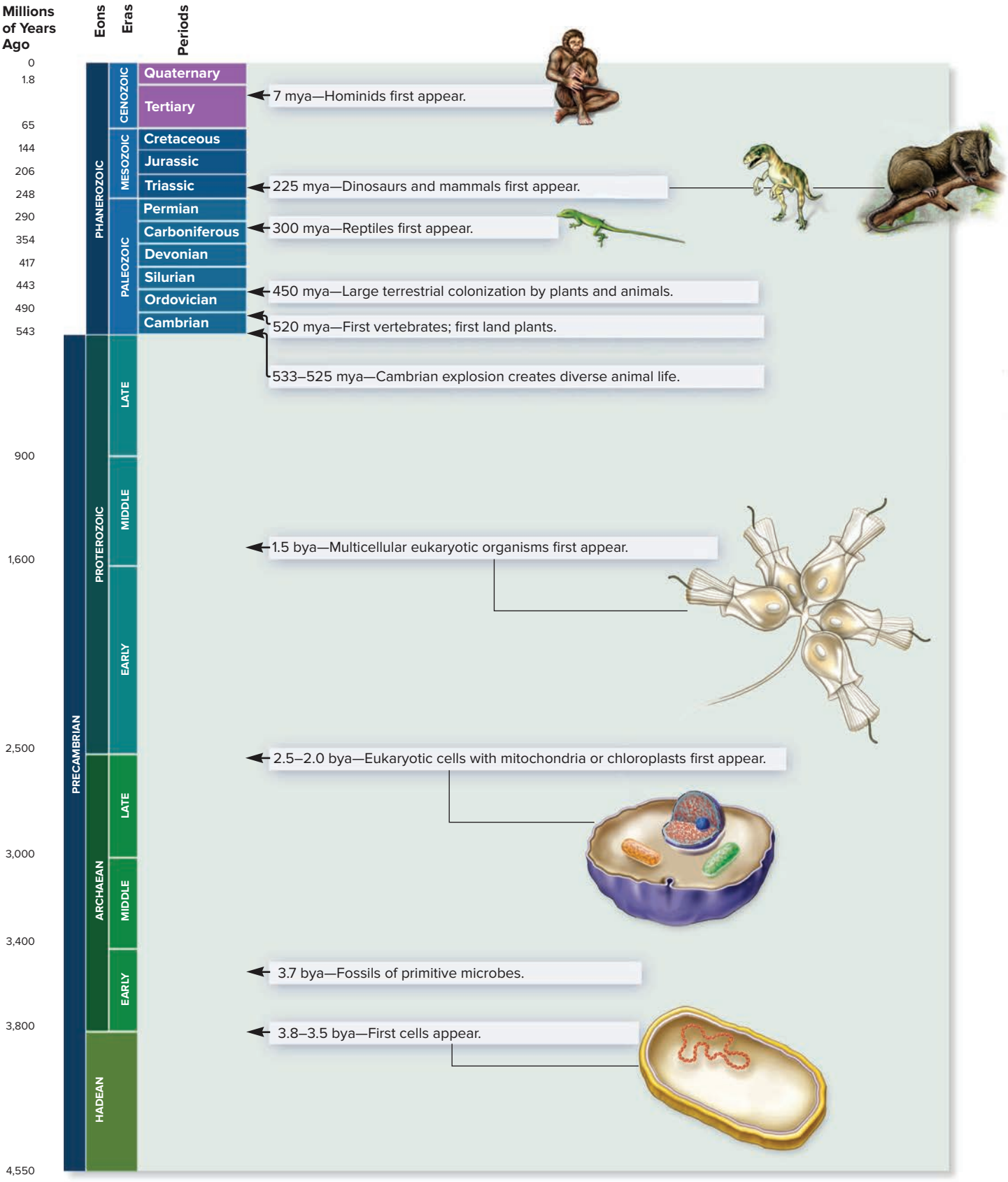


Figure 1.4 An Overview of the History of Life on Earth. mya = million years ago; bya = billion years ago.

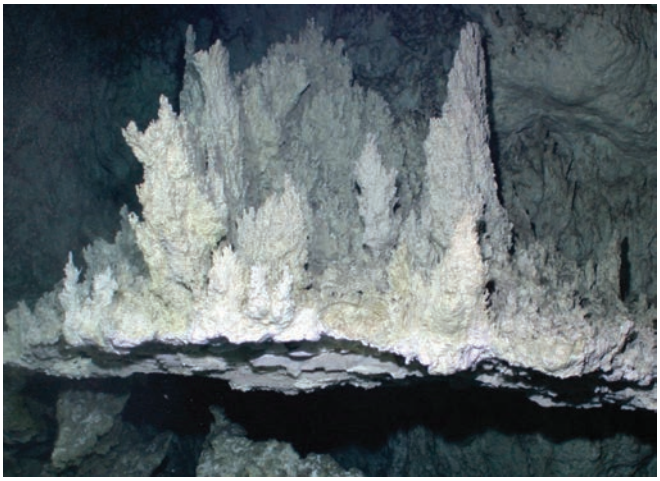
MICROBIAL DIVERSITY & ECOLOGY

1.1 Hydrothermal Vents: Did Life Begin Under the Sea?

Whether or not early life was RNA-based, one thing is clear: the origin of life needed energy to synthesize biomolecules. So, perhaps the most fundamental evolutionary question is "Where did biomolecules and the energy needed to build them come from?" Three hypotheses have been suggested. First, the *panspermia theory* speculates that meteorites bombarded our planet, bringing with them other-worldly biomolecules. Second, the more familiar *primordial soup theory* suggests that organic molecules were spontaneously assembled by an input of energy, such as lightning strikes. The last theory, which has gained evidence in recent years, hypothesizes that both the energy and the molecules originated in hydrothermal vents. Let's explore the *hydrothermal vent theory.*

Hydrothermal vents are geothermally active deep-sea chasms thousands of meters below the surface of the ocean. Their discovery in 1977 sparked tremendous excitement as images of entirely new ecosystems with mysterious organisms captured the attention of scientists and the public (*see section 27.2*). These vents pump 400°C sulfide-rich water into cold ambient water, causing the sulfide to instantly precipitate, so these chimneylike structures are dubbed "black smokers." In 2000, scientists made yet another deep-sea discovery with a different kind of vent system. These are cooler (45–90°C) and alkaline (pH 9–11). When these waters mix with the surrounding seawater (pH about 8.0), calcium carbonate precipitates, forming white chimneys, as seen in the Lost City vents (**box figure**).

This pH gradient is critical to the hypothesis that a vent system, such as Lost City, could be the origin of biomolecules. As you may have learned when studying mitochondria or batteries, the separation of positive and negative charges captures potential energy (remember that energy can't be created). In Lost City vents, the thin walls of the chimneys serve to separate these fluids with as much as a 3-unit pH difference. The question now being asked is "Was this potential energy tapped to convert CO_2 in seawater to simple carbon-based molecules, such as amino acids, short hydrocarbons, and others?"



D.Kelley, University of Washington.

If the answer is yes, a 2019 study shows that a mixture of molecules called single-chain amphiphiles (SCAs), which are simpler versions of more familiar phospholipids, can form vesicles in hot, alkaline pH seawater that mimics that of Lost City. Putting this together, we can hypothesize a series of events that occurred 3.7–4.0 billion years ago. First, the presence of the pH gradient across geological barriers in the Lost City drove the formation of random organic molecules, some of which were SCAs. These SCAs accumulated and formed vesicles that entrapped fluids preserving the pH gradient. These vesicles had the energy to test the formation of different molecules. Was one of them RNA?

(tRNA) to construct proteins. Also rRNA itself catalyzes peptide bond formation during protein synthesis. Thus RNA seems to be well poised for its importance in the development of proteins (figure 1.5). Because RNA and DNA are structurally similar, RNA could have given rise to double-stranded DNA. It is suggested that once DNA evolved, it became the storage facility for genetic information because it provides a more chemically stable structure. Two other pieces of evidence support the RNA world hypothesis: the fact that the energy currency of cells, ATP, is a ribonucleotide and the discovery that RNA can regulate gene expression. ▶| *ATP: the major energy currency of cells (section 10.2); Riboswitches: effector-mRNA interactions regulate transcription (section 14.3); Translational riboswitches (section 14.4)*

However, the RNA world hypothesis is not without problems, and more recent experiments suggest the first nucleic acids may have been a mix of DNA and RNA molecules. Another area of research also fraught with considerable debate is the evolution of metabolism. The early Earth was a hot environment that lacked oxygen. Thus cells that arose there must have been able to use the available energy sources under these harsh conditions. Today there are heat-loving archaea capable of using inorganic molecules such as FeS as a source of energy. Some suggest that this interesting metabolic capability is a remnant of

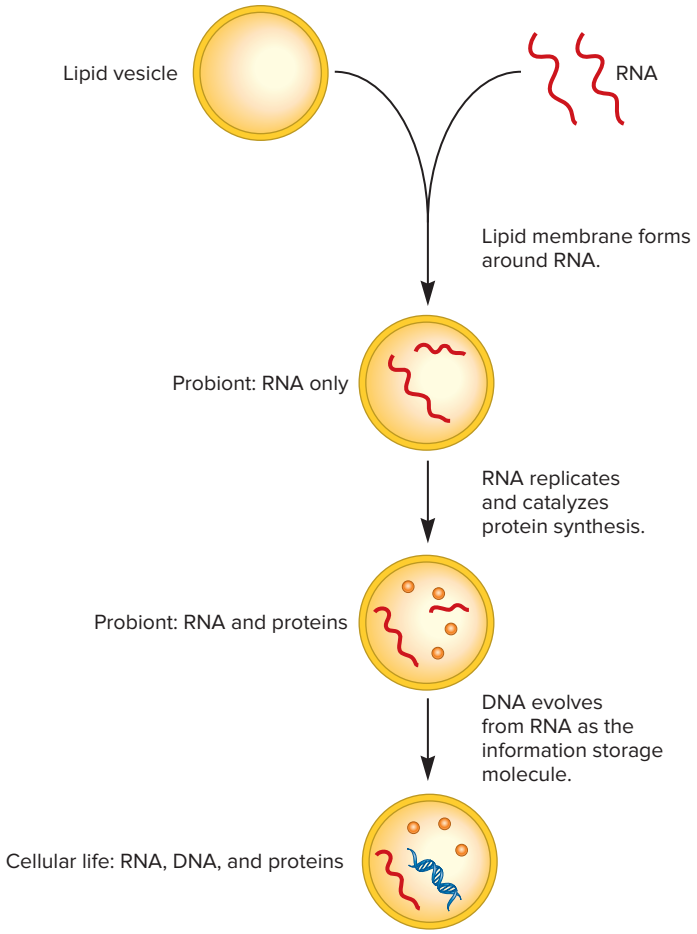


Figure 1.5 The RNA World Hypothesis for the Origin of Life.

MICRO INQUIRY *Why are the probionts pictured above not considered cellular life?*

the first form of energy metabolism. Another metabolic strategy, oxygen-releasing photosynthesis (oxygenic photosynthesis), appears to have evolved perhaps as early as 2.7 billion years ago. This is supported by the discovery of ancient stromatolites (**figure 1.6**). Stromatolites are layered rocks formed by the incorporation of mineral sediments into layers of cyanobacteria growing in thick mats on surfaces. The oxygen released by these early cyanobacteria is thought to have altered Earth's atmosphere to its current oxygen-rich state, allowing the evolution of additional energy-capturing strategies such as aerobic respiration, the oxygen-consuming metabolic process used by many microbes and animals.

Evolution of the Three Domains of Life

Look closely at figure 1.3 and find a line labeled "Origin." This is where data indicate the ***l*ast *u*niversal *c*ommon *a*ncestor (LUCA)** to all three domains should be placed. LUCA is the most recent organism from which all three types of life—bacterial, archaeal, and eukaryotic—arose. On this tree of life,



Figure 1.6 Stromatolites. Modern stromatolites from Western Australia. Each stromatolite is a rocklike structure, typically 1 m in diameter, containing layers of cyanobacteria.

Horst Mahr/imagebroker/age fotostock

LUCA is on the bacterial branch, which means that Archaea and Eukarya evolved independently, separate from Bacteria.

The evolutionary relationship of Archaea and Eukarya is still a matter of considerable debate. According to the **universal phylogenetic tree** (figure 1.3), Archaea and Eukarya shared common ancestry but diverged and became separate domains. Recent evidence supports the notion that Eukarya evolved from Archaea *(see Microbial Diversity & Ecology 26.1)*. The close evolutionary relationship of these two forms of life is still evident in the manner in which they process genetic information. For instance, certain protein subunits of archaeal and eukaryotic RNA polymerases, the enzymes that catalyze RNA synthesis, resemble each other to the exclusion of those of bacteria. However, archaea have other features that are most similar to their counterparts in bacteria (e.g., mechanisms for conserving energy). This has further complicated and fueled the debate. The evolution of the nucleus and endoplasmic reticulum is also controversial. However, hypotheses regarding the evolution of other membrane-bound organelles are more widely accepted and are considered next.

Mitochondria, Mitochondria-Like Organelles, and Chloroplasts Evolved from Endosymbionts

The **endosymbiotic hypothesis** is generally accepted as the origin of several eukaryotic organelles, including mitochondria, chloroplasts, and hydrogenosomes. **Endosymbiosis** is an interaction between two organisms in which one organism lives inside the other. The original endosymbiotic hypothesis proposed that over time a bacterial endosymbiont of an ancestral cell in the eukaryotic lineage lost its ability to live independently. If the intracellular bacterium used aerobic respiration, it became a mitochondrion. If the endosymbiont was a cyanobacterium and therefore performed photosynthesis, it became a chloroplast (**figure 1.7**).

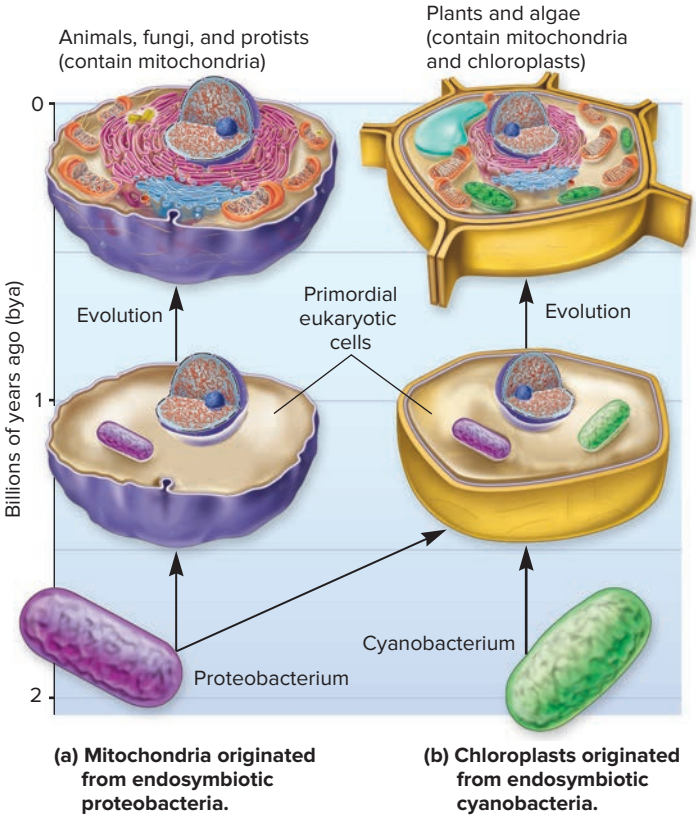


Figure 1.7 The Endosymbiotic Theory. (a) According to this hypothesis, mitochondria derived from a bacterium in the phylum Proteobacteria. (b) A similar phenomenon occurred for chloroplasts, which derived from cyanobacteria.

Although the mechanism by which the endosymbiotic relationship was established is unknown, there is considerable evidence to support this hypothesis. Mitochondria and chloroplasts contain DNA and ribosomes; both are similar to bacterial DNA and ribosomes. Peptidoglycan, the unique bacterial cell wall molecule, has even been found between the two membranes that enclose the chloroplasts of some algae. Indeed, inspection of figure 1.3 shows that both organelles belong to the bacterial lineage. More specifically, mitochondria are most closely related to bacteria called proteobacteria. The chloroplasts of plants and green algae are thought to have descended from an ancestor of the cyanobacterial genus *Prochloron,* which contains species that live within marine invertebrates. ▶| *Phylum Cyanobacteria: oxygenic photosynthetic bacteria (section 20.4); The proteobacterial origin of mitochondria (section 21.1)*

The endosymbiotic hypothesis for mitochondria has been refined by the **hydrogen hypothesis.** This asserts that the endosymbiont was an anaerobic bacterium that produced H_2 and CO_2 as end products of its metabolism. Over time, the host became dependent on the H_2 produced by the endosymbiont. Ultimately the endosymbiont evolved into one of several organelles (*see figure 5.13*). Some endosymbionts evolved into organelles such as a hydrogenosome—an organelle found in some extant protists that produces ATP by a process called fermentation *(see figure 5.15).*

Evolution of Cellular Microbes

Although the history of early cellular life forms may never be known, we know that once microbial cells arose, they were subjected to the same evolutionary processes as modern organisms. The ancestral bacteria, archaea, and eukaryotes possessed genetic information that could be duplicated, lost, or mutated in other ways. Mutations could have many outcomes. Some led to the death of the microbe, but others allowed new functions and characteristics to evolve. Mutations that allowed the organism to increase its rate of reproduction or survival were selected and passed on to subsequent generations. In addition to selective forces, geographic isolation of populations allowed some groups to evolve separately from others. Thus selection and isolation led to the eventual development of new collections of genes (i.e., genotypes) and new species.

In addition to mutation, other mechanisms exist for reconfiguring genomes and therefore creating genetic diversity. Most eukaryotic species increase their genetic diversity by reproducing sexually, whereby each offspring has a mixture of parental genes and a unique genotype. Bacteria and archaea do not reproduce sexually. They increase their genetic diversity by mutation and horizontal gene transfer (HGT). During HGT, genetic information from a donor organism is transferred to a recipient, creating a new genotype in the recipient. In this way genetic information is passed between individuals of the same generation and even between species found in different domains of life. Genome sequencing has revealed that HGT has played an important role in the evolution of all microbial species. Importantly, HGT still occurs in bacteria and archaea leading to the rapid evolution of microorganisms with antibiotic resistance, new virulence properties, and novel metabolic capabilities. The outcome of HGT is that most microbes have mosaic genomes composed of bits and pieces of the genomes of other organisms. ▶| *Horizontal gene transfer: creating genetic variation the asexual way (section 16.4)*

Phylogenetic or **phyletic classification systems** compare organisms on the basis of evolutionary relationships. The term **phylogeny** (Greek *phylon,* tribe or race; *genesis,* generation or origin) refers to the evolutionary development of organisms. As discussed, microbial phylogeny relies on comparisons of multiple features found in extant organisms. These include cell wall structure, biomolecules such as fatty acids, and certain housekeeping proteins (proteins used to maintain cellular life, therefore found in many different organisms), and nucleotide sequences, particularly of small subunit rRNA molecules (SSU rRNA) (**table 1.1**). ▶| *Bacterial ribosomes (section 3.7); Archaeal ribosomes (section 4.3); Eukaryotic ribosomes (section 5.5)*

Phylogenetic Trees

Figure 1.3 is an example of a **phylogenetic tree.** The goal of phylogenetic tree construction is to display the evolutionary

Table 1.1 Comparison of Bacteria, Archaea, and Eukarya

Property	Bacteria	Archaea	Eukarya
Organization of Genetic Material			
Membrane-bound nucleus	No	No	Yes
DNA complexed with histones	No	Some	Yes
Chromosomes	Usually one circular chromosome; chromosomes have single origin of replication; some are polyploid	One circular chromosome; some have chromosomes with multiple origins of replication; some are polyploid	Multiple, linear chromosomes with multiple origins of replication; usually diploid
Plasmids	Yes	Yes	Rare
Introns in genes	Rare	Rare	Yes
Nucleolus	No	No	Yes
Mitochondria, Chloroplasts, Endoplasmic Reticulum, Golgi, and Lysosomes Observed	No	No	Yes
Plasma Membrane Lipids	Ester-linked phospholipids; some have sterols	Glycerol diethers and diglycerol tetraethers	Ester-linked phospholipids and sterols
Flagella	Submicroscopic in size; filament composed of single type of flagellin protein	Submicroscopic in size; filaments composed of more than one type of archaellin protein	Microscopic in size; membrane bound; usually 20 microtubules in 9 + 2 pattern
Peptidoglycan in Cell Walls	Yes	No	No
Ribosome Size and Structure	70S; 3 rRNAs; 49–59 ribosomal proteins	70S; 3 rRNAs; 58–68 ribosomal proteins	80S; 4 rRNAs; 78–80 ribosomal proteins
Cytoskeleton	Rudimentary	Rudimentary	Yes

relationships between different organisms. Trees are built by comparing the sequences of amino acids or nucleotides from diverse organisms. Amino acid sequences are often compared because nucleotide changes may not alter the resulting protein and therefore have little or no impact on evolution. Historically, the rRNAs from small ribosomal subunits (16S from bacteria and archaea and 18S from eukaryotes) were the molecules of choice for inferring microbial phylogenies and making taxonomic assignments for several reasons. Like housekeeping proteins (proteins with essential functions), SSU rRNAs play the same role in all microorganisms and are absolutely necessary for survival. Therefore neither housekeeping proteins nor genes encoding SSU rRNAs tolerate large changes in sequence. The utility of SSU rRNAs is extended by the presence of certain sequences within SSU rRNA genes that vary among organisms as well as other regions that are quite similar. The variable regions enable comparison between closely related microbes, whereas the stable sequences allow the comparison of more distantly related microorganisms. The stability of SSU rRNA sequences means that they evolve very slowly, so analysis of housekeeping genes is more frequently used to infer the evolution of more closely related organisms.

Comparative analysis of SSU rRNA sequences from thousands of organisms has demonstrated the presence of **oligonucleotide signature sequences** (**figure 1.8**). These are short, conserved nucleotide sequences specific for phylogenetically defined groups of organisms. Thus the signature sequences found in bacterial rRNAs are rarely or never found in archaeal rRNAs, and vice versa. Likewise, the 18S rRNA of eukaryotes bears signature sequences that are specific to the domain Eukarya.

Approaches to building a phylogenetic tree can be divided into two broad categories: a distance-based approach and a character-based approach. Distance-based approaches are the most intuitive. Here the differences between the aligned sequences are counted for each pair and summarized in a single statistic (**figure 1.9**). This value serves as a measure of the evolutionary distance between the organisms; the more differences counted,

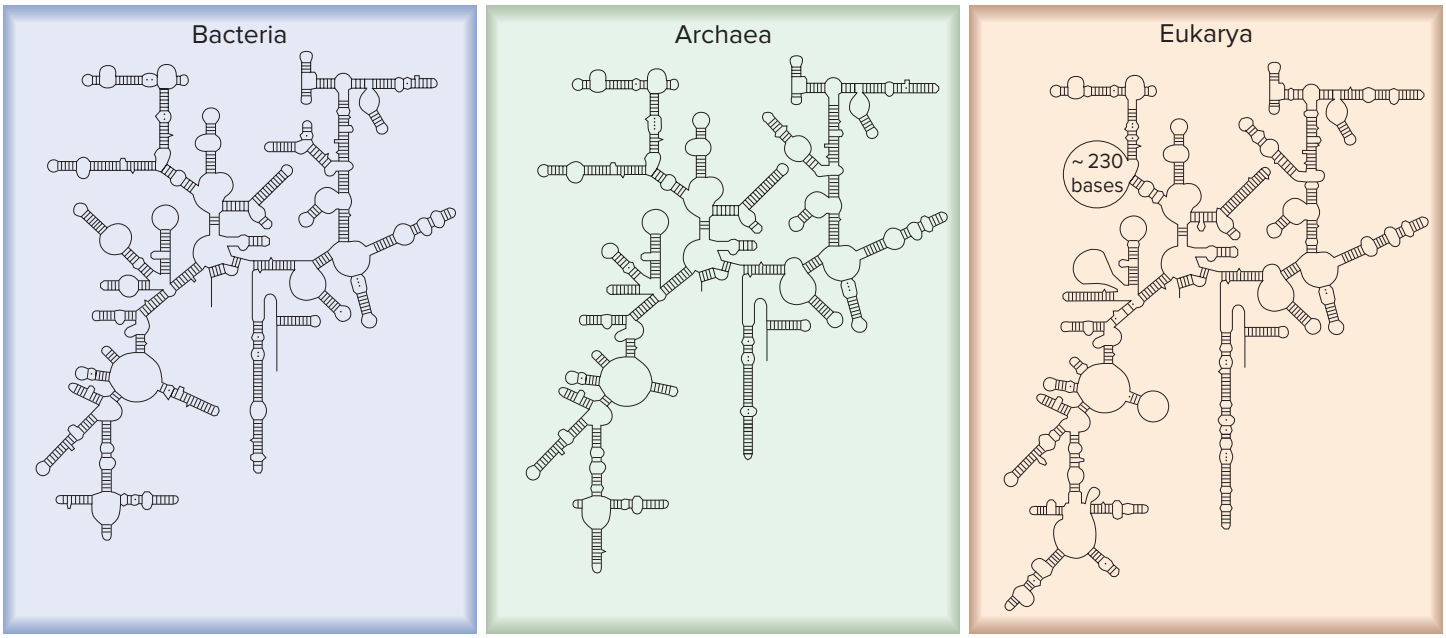


Figure 1.8 Signature rRNA Help Identify Microbes. Representative examples of rRNA secondary structures from the three domains: Bacteria *(Escherichia coli),* Archaea *(Methanococcus vannielii),* and Eukarya *(Saccharomyces cerevisiae).*

the greater the evolutionary distance. The evolutionary distances from many comparisons are used by sophisticated computer programs to construct the tree. The tip of each branch in the tree (called a *node*) represents one of the organisms used in the comparison. The distance from one node to another is the evolutionary distance between the two organisms.

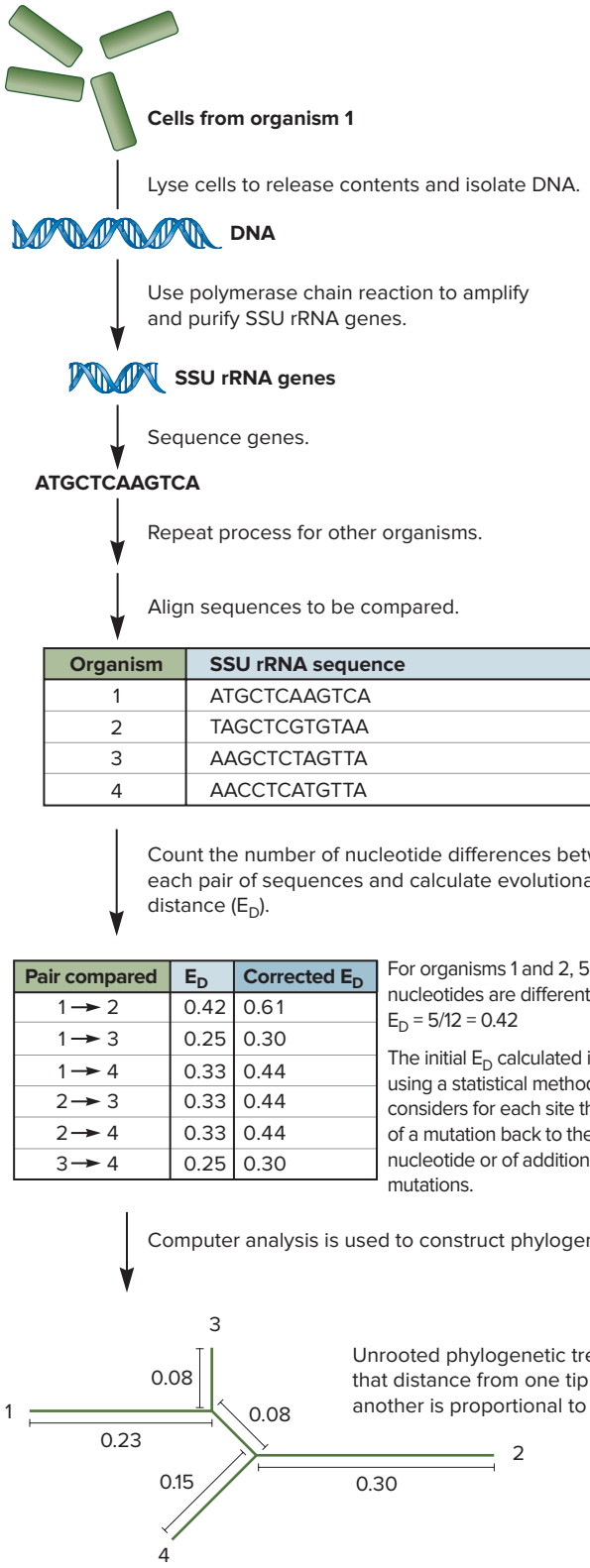
An algorithm analyzes this information to generate a tree. One algorithm called cluster analysis serially links pairs that are ever more distantly related (i.e., start with those with the least number of sequence differences and move to those with the most). Although this is the easiest to understand, it can generate trees based on artifact. Neighbor joining is another distance-based method that uses a different matrix to modify the distance between each pair of nodes based on the average divergence from all other nodes.

Character-based approaches for phylogenetic tree building are more complicated but generate more robust trees. These methods start with assumptions about the pathway of evolution, infer the ancestor at each node, and choose the best tree according to a specific model of evolutionary change. Sometimes called tree-searching, these methods include maximum parsimony, which assumes that the fewest number of changes occurred between ancestor and extant organisms. Another approach is called maximum likelihood. This requires a large data set, because for each possible tree that can be built, its probability (i.e., the likelihood) based on certain evolutionary and molecular information is determined so that the tree with the greatest probability based on these criteria is selected. Bayesian inference is another character-based approach. Rather than looking at a single tree, Bayesian inference analyzes multiple potential trees and calculates the probability that each branch would appear based on this comparison.

Once a tree is constructed, it is important to get a sense of whether the placement of its branches and nodes is legitimate. There are a variety of methods to assess the strength of a tree, but the most common is bootstrapping. Bootstrapping involves re-analyzing a randomly selected subset of the data presented on the tree. A bootstrap value is the percent of times in which that particular branch was found. Typically, bootstrap values of 70% or greater are thought to support a tree. Of note is that Bayesian inference values are also reported as percentages, but they are not directly comparable to bootstrap values. Only values greater than 95% are acceptable when Bayesian inference is used.

Two things should be kept in mind when examining phylogenetic trees. The first is that they are molecular trees, not organismal trees. In other words, they represent, as accurately as possible, the evolutionary history of a molecule (e.g., rRNA). Second, the distance between nodes is a measure of relatedness, not of time. If the distance along the lines is very long, then the two organisms are more evolutionarily diverged (i.e., less related). However, we do not know when they diverged from each other. This concept is analogous to a printed map that accurately shows the distance between two cities but because of many factors (traffic, road conditions, etc.) cannot show the time needed to travel that distance.

Importantly, a tree may be unrooted or rooted. An unrooted tree (**figure 1.10*a***) represents phylogenetic relationships but does not indicate which organisms are more primitive relative to the others. Figure 1.10*a* shows that A is more closely related to C than it is to either B or D, but it does not indicate which of the



Organism	SSU rRNA sequence
1	ATGCTCAAGTCA
2	TAGCTCGTGTAA
3	AAGCTCTAGTTA
4	AACCTCATGTTA

Pair compared	E_D	Corrected E_D
1 → 2	0.42	0.61
1 → 3	0.25	0.30
1 → 4	0.33	0.44
2 → 3	0.33	0.44
2 → 4	0.33	0.44
3 → 4	0.25	0.30

Figure 1.9 The Construction of a Phylogenetic Tree Using a Distance Method. The polymerase chain reaction is described in chapter 17.

MICRO INQUIRY *Why does the branch length indicate amount of evolutionary change but not the time it took for that change to occur?*

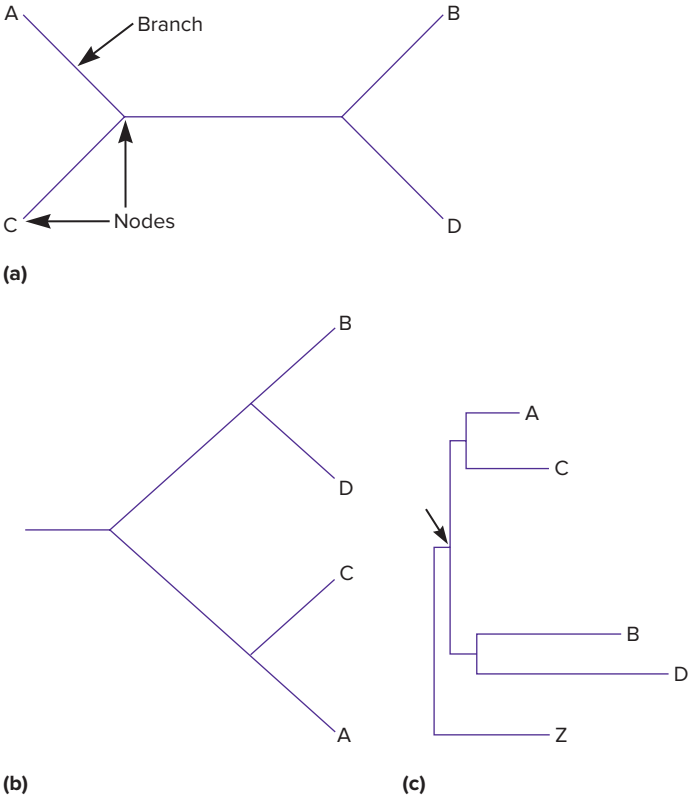


Figure 1.10 Phylogenetic Tree Topologies. (a) Unrooted tree joining four taxonomic units (A, B, C, D). (b) Rooted tree. (c) The tree can be rooted by adding an outgroup, represented by Z. The arrow indicates a speciation event that resulted in the development of new species from an ancestral organism.

four species might be the oldest. In contrast, the rooted tree (figure 1.10*b*) includes a node (taxonomic unit) that serves as the common ancestor and shows the development of the four species from this root. It is much more difficult to develop a rooted tree. For example, there are 15 possible rooted trees that connect four species but only three possible unrooted trees.

An unrooted tree can be rooted by adding data from an outgroup—a species known to be very distantly related to all the species in the tree (figure 1.10*c*). The root is determined by the point of the tree where the outgroup joins. This provides a point of reference to identify the oldest node on the tree, which is the node closest to the outgroup. So, for example, in figure 1.10*c*, organism Z is the outgroup and the oldest node on the tree is marked with an arrow.

Microbial Taxonomy

The science of classifying living things is called **taxonomy.** Taxonomy consists of three separate but interrelated parts: classification, nomenclature (naming), and identification. A taxonomic scheme is used to arrange organisms into groups called taxa (s., **taxon**) based on mutual similarity. Microbes are placed in taxonomic levels arranged in a nonoverlapping hierarchy so that each level includes not only the traits that define the rank above it but also a new set of more restrictive

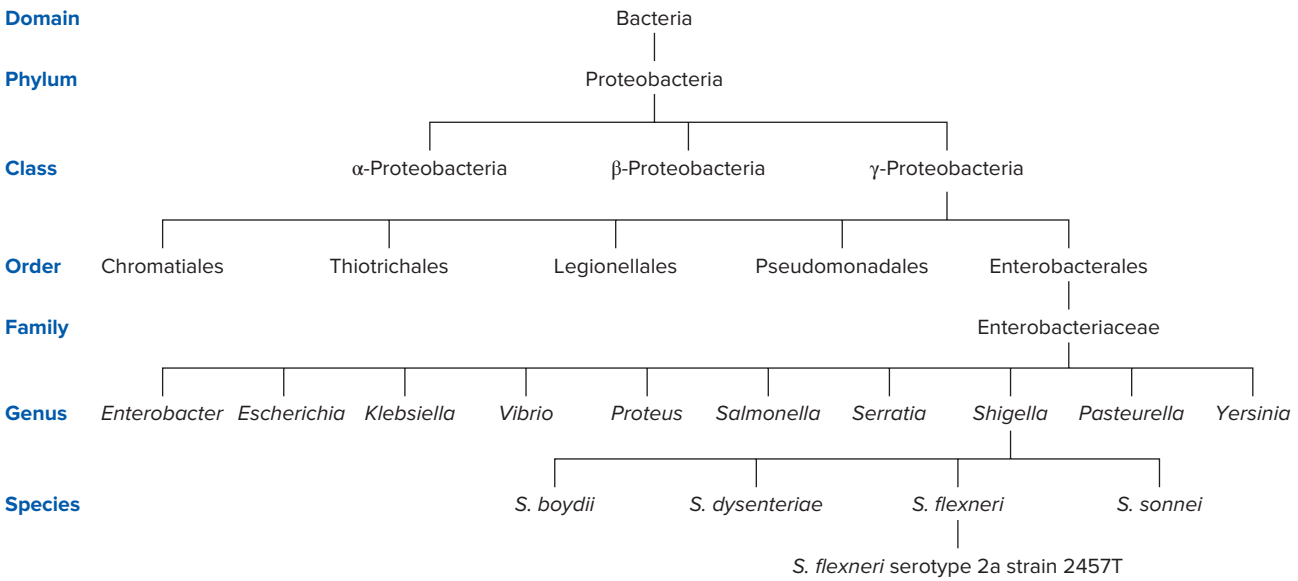


Figure 1.11 Hierarchical Arrangement in Taxonomy. In this example, members of the genus *Shigella* are placed within higher taxonomic ranks. Not all classification possibilities are given for each rank to simplify the diagram. Note that *-ales* denotes order and *-ceae* indicates family.

traits (**figure 1.11**). Thus within each domain—Bacteria, Archaea, or Eukarya—each organism is assigned (in descending order) to a phylum, class, order, family, genus, and species epithet or name. Some microbes are also given a subspecies designation. Microbial groups at each level have a specific suffix that indicates rank or level.

Microbiologists name microbes using the binomial system of the eighteenth-century biologist and physician Carl Linnaeus. The Latin, italicized name consists of two parts. The first part, which is capitalized, is the generic name (i.e., the name of the genus to which the microbe belongs), and the second is the uncapitalized species epithet. For example, the bacterium that causes plague is called *Yersinia pestis*. Often the name of an organism will be shortened by abbreviating the genus name with a single uppercase letter (e.g., *Y. pestis*).

This straightforward organizational approach is complicated by the fact that bacteria and archaea do not reproduce sexually. You may recall from a general biology class that plant and animal **species** are defined as a group of interbreeding or potentially interbreeding natural populations reproductively isolated from other groups. This definition also is appropriate for the many eukaryotic microbes that reproduce sexually. However, bacterial and archaeal species cannot be defined by this criterion because they do not reproduce sexually. Therefore comparisons of genome sequences are often used to distinguish one species from another. An appropriate definition is currently debated. A common definition is that bacterial and archaeal species are a collection of strains that share many stable properties and differ significantly from other groups of strains. A **strain** consists of the descendants of a single, pure microbial culture. Strains within a species may be described in a number of different ways. Biovars are variant strains characterized by biochemical or physiological differences, morphovars differ morphologically, serovars have distinctive properties that can be detected by antibodies, and pathovars are pathogenic strains distinguished by the plants in which they cause disease.

Although microbiologists continue to use Linnaeus's classification system, the ongoing explosion in metagenomic analysis has had an impact on this historically accepted hierarchy. Within the last few decades, thousands of 16S rRNA genes and protein-coding genes have been sequenced that do not belong to any previously defined taxa. These data have led to the recent development of the taxonomic classification **superphylum,** below domain and above phylum (e.g., in figure 1.11, superphylum would be placed between Bacteria and Proteobacteria). A superphylum includes organisms of several phyla that share a number of distinctive characteristics, such as unusual morphological or metabolic features.

Comprehension Check

1. Describe two reasons why RNA is thought to be the first self-replicating biomolecule.
2. Explain the endosymbiotic hypothesis of the origin of mitochondria, hydrogenosomes, and chloroplasts. List two pieces of evidence that support this hypothesis.
3. What is the goal of phylogenetic tree construction?
4. In general terms, what is the difference between a distance-based and character-based tree? Does a rooted or unrooted phylogenetic tree provide more information? Explain your answer.
5. What is the difference between mutation and horizontal gene transfer?
6. What is the correct way to write this microbe's name: *bacillus subtilis,* Bacillus subtilis, *Bacillus Subtilis,* or *Bacillus subtilis?* Identify the genus name and the species epithet.

1.3 Microbiology Advanced as New Tools for Studying Microbes Were Developed

After reading this section, you should be able to:

- **a.** Evaluate the importance of the contributions made by Hooke, Leeuwenhoek, Pasteur, Lister, Koch, Beijerinck, von Behring, Kitasato, Metchnikoff, and Winogradsky to microbiology
- **b.** Outline a set of experiments that might be used to decide if a particular microbe is the causative agent of a disease
- **c.** Predict the difficulties that might arise when using Koch's postulates to determine if a microbe causes a disease unique to humans

Even before microorganisms were seen, some investigators suspected their existence and role in disease. Among others, the Roman philosopher Lucretius (about 98–55 BCE) and the physician Girolamo Fracastoro (1478–1553) suggested that disease was caused by invisible living creatures. However, until microbes could actually be seen and studied in some other way, their existence remained a matter of conjecture. Therefore **microbiology** is defined not only by the organisms it studies but also by the tools used to study them. The development of microscopes was the critical first step in the evolution of the discipline. However, microscopy alone is unable to answer the many questions scientists ask about microbes. A distinct feature of microbiology is that microorganisms are usually removed from their normal habitats and grown in isolation, apart from all other microbes. This is called a **pure** or **axenic culture.** Although the development of techniques for isolating microbes in pure culture was another critical step in microbiology's history, it is now recognized as having limitations. Microbes in pure culture are in some ways like animals in a zoo; just as a zoologist cannot fully understand animals by studying them in zoos, microbiologists cannot fully understand microbes by studying them in pure culture. Today molecular genetic techniques and genomic analyses are providing new insights into the lives of microbes.

Here we describe how the tools used by microbiologists have influenced the development of the field. As microbiology evolved as a science, it contributed greatly to the well-being of humans. The historical context of some of the important discoveries in microbiology is shown in **figure 1.12.**

Microscopy Led to the Discovery of Microorganisms

The earliest microscopic observations of organisms appear to have been made between 1625 and 1630 on bees and weevils by the Italian Francesco Stelluti (1577–1652), using a microscope probably supplied by Galileo (1564–1642). Robert Hooke (1635–1703) is credited with publishing the first drawings of microorganisms in the scientific literature. In 1665 he published a highly detailed drawing of the fungus *Mucor* in his book *Micrographia. Micrographia* is important not only for its exquisite drawings but also for the information it provided on building microscopes. One design discussed in *Micrographia* was probably a prototype for the microscopes built and used by the amateur microscopist Antony van Leeuwenhoek (1632–1723) of Delft, the Netherlands. Leeuwenhoek earned his living selling men's clothing and accessories but spent much of his spare time constructing simple microscopes composed of double convex glass lenses held between two silver plates (**figure 1.13*a***). His microscopes could magnify about 50 to 300 times, and he may have illuminated his liquid specimens by placing them between two pieces of glass and shining light on them at a 45-degree angle to the specimen plane. This would have provided a form of dark-field illumination whereby organisms appeared as bright objects against a dark background. Beginning in 1673, Leeuwenhoek sent detailed letters describing his discoveries to the Royal Society of London. It is clear from his descriptions that he saw both bacteria and "protists" (figure 1.13*b*). ▶| *Dark-field microscope: bright object, dark background (section 2.2)*

Culture-based Methods for Studying Microorganisms Were a Major Development

As important as Leeuwenhoek's observations were, the development of microbiology essentially languished for the next 200 years until techniques for isolating and culturing microbes in the laboratory were formulated. During this time, scientists grappled with the conflict over the theory of spontaneous generation. This conflict and the subsequent studies on the role played by microorganisms in causing disease ultimately led to what is now called the golden age of microbiology.

Spontaneous Generation

From earliest times, people had believed in **spontaneous generation**—that living organisms could develop from nonliving matter. This view was challenged by the Italian physician Francesco Redi (1626–1697), who carried out a series of experiments on decaying meat, which was thought to produce maggots spontaneously. Using covered and uncovered containers of meat, Redi clearly demonstrated that maggots on decaying meat resulted from the presence of fly eggs, and meat did not spontaneously generate maggots. Other experiments helped discredit the theory for larger organisms. However, Leeuwenhoek's communications on microorganisms renewed the controversy. Some proposed that microbes arose by spontaneous generation but larger organisms did not. They pointed out that boiled extracts of hay or meat gave rise to microorganisms after sitting for a while. In 1748, the English priest John Needham (1713–1781) suggested that the organic matter in these extracts contained a "vital force" that could confer the properties of life on nonliving matter.

A few years after Needham's experiments, the Italian priest and naturalist Lazzaro Spallanzani (1729–1799) sealed glass flasks that contained water and seeds and then placed the flasks in boiling water for about 45 minutes. He found that no growth took place as long as the flasks remained sealed. He proposed